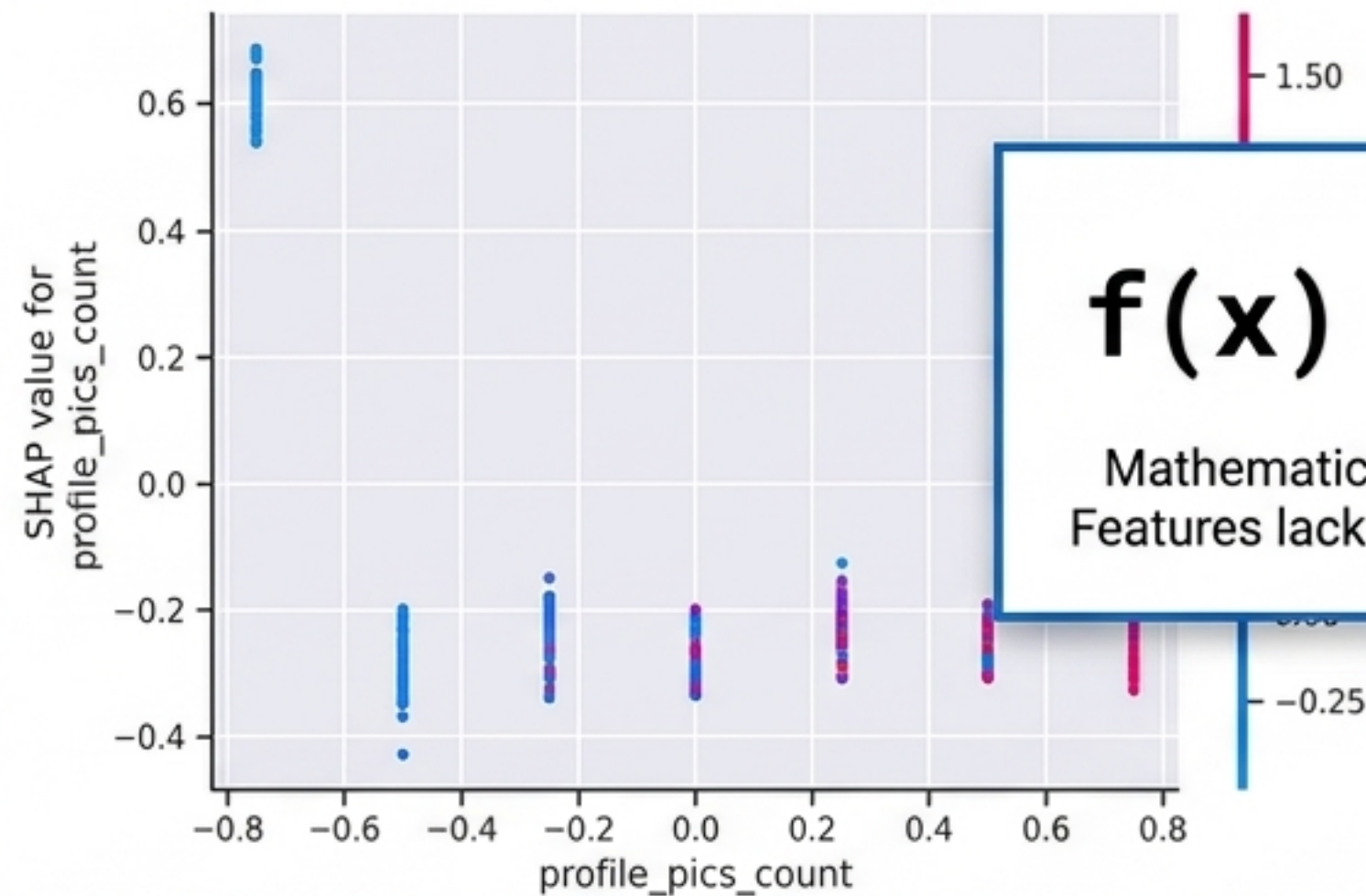


Global Impact: SHAP Attributions

SHAP Interaction Analysis: profile_pics_count x profile_completeness

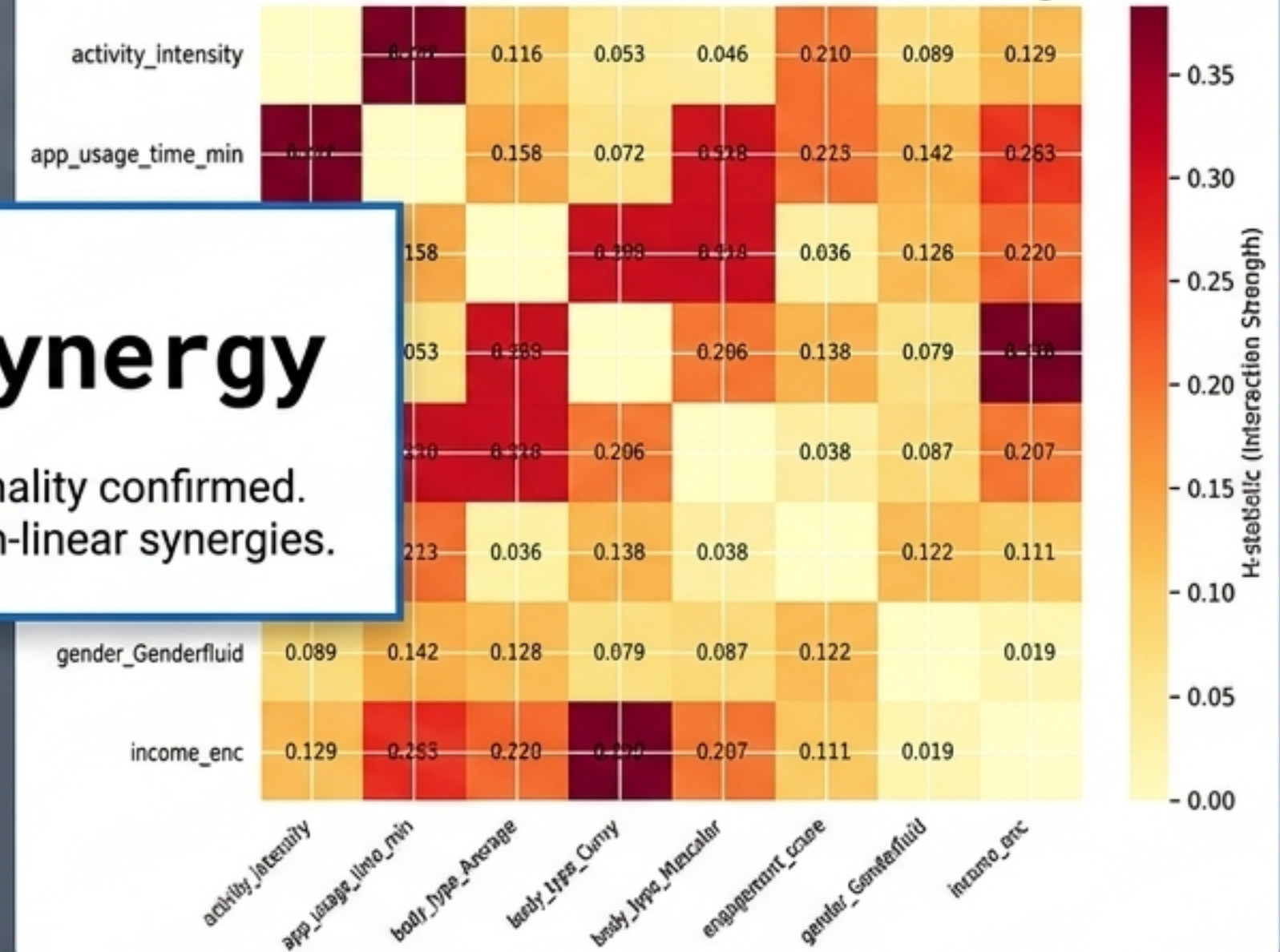


$f(x) \neq \text{Synergy}$

Mathematical orthogonality confirmed.
Features lack hidden non-linear synergies.

Synergy Mapping: Friedman's H-Statistic

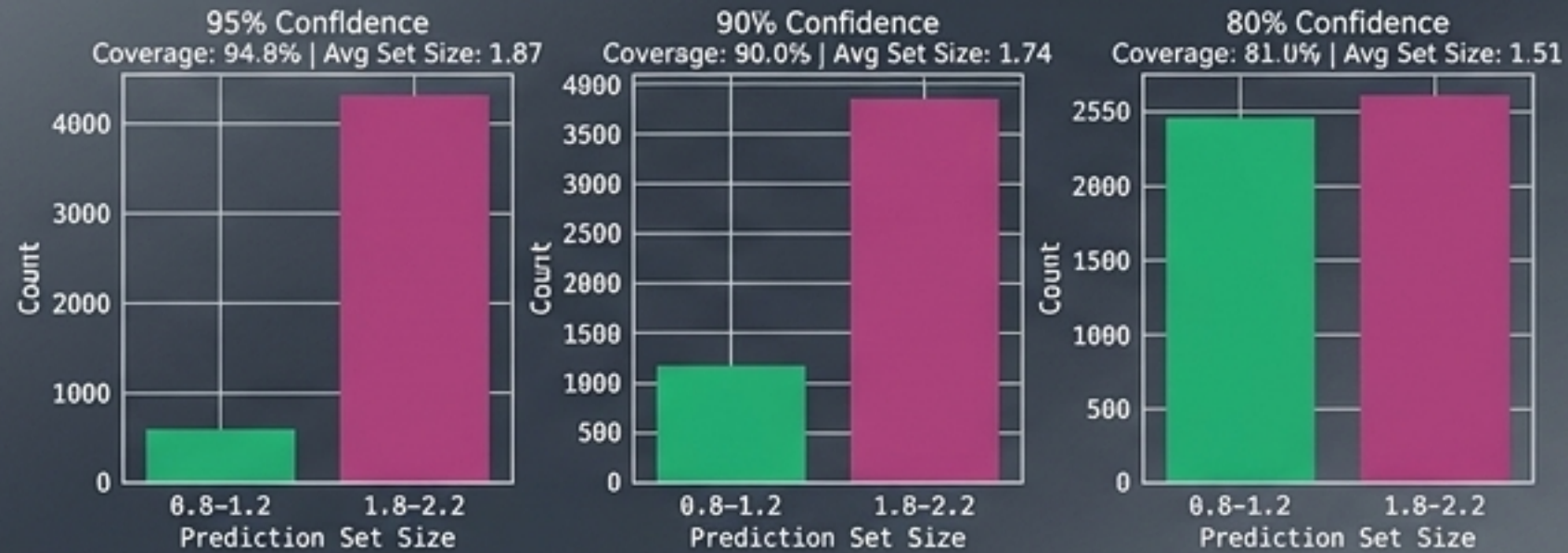
Friedman's H-Statistic: Pairwise Feature Interaction Strength



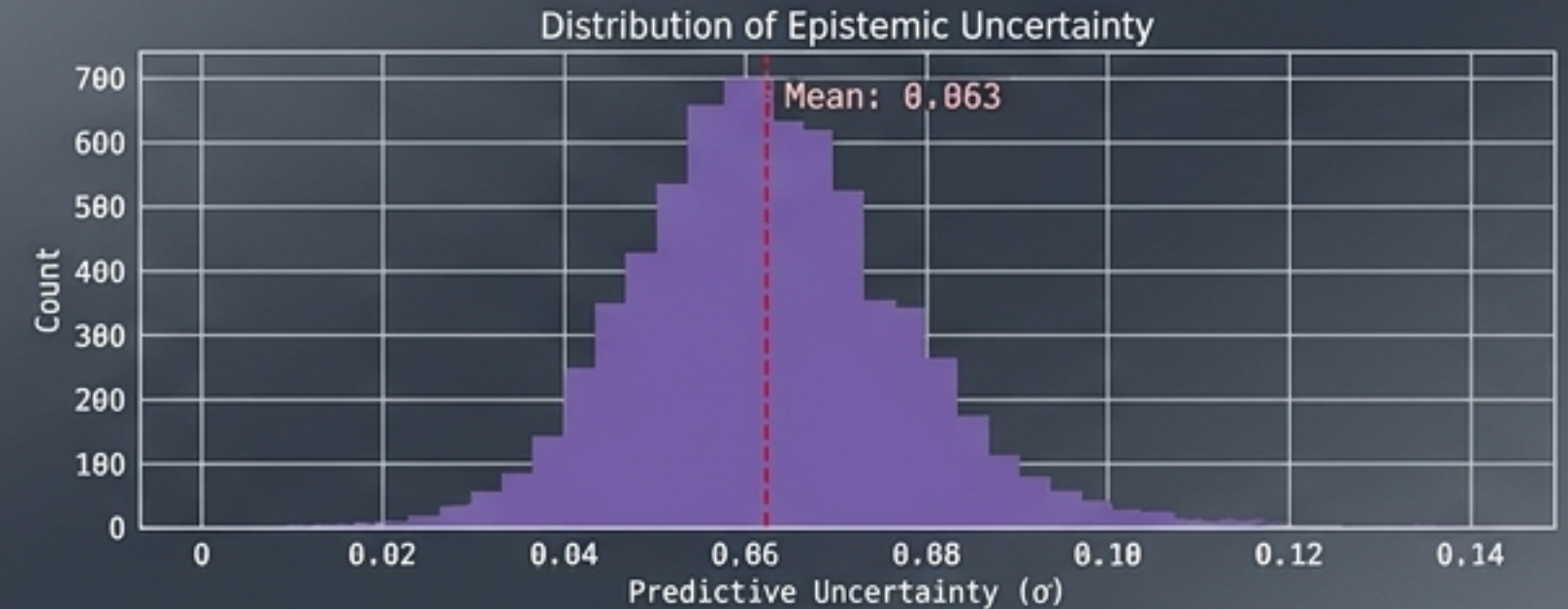
Trustworthy AI: Uncertainty & Fairness

Mathematical Bounding & Epistemic Uncertainty

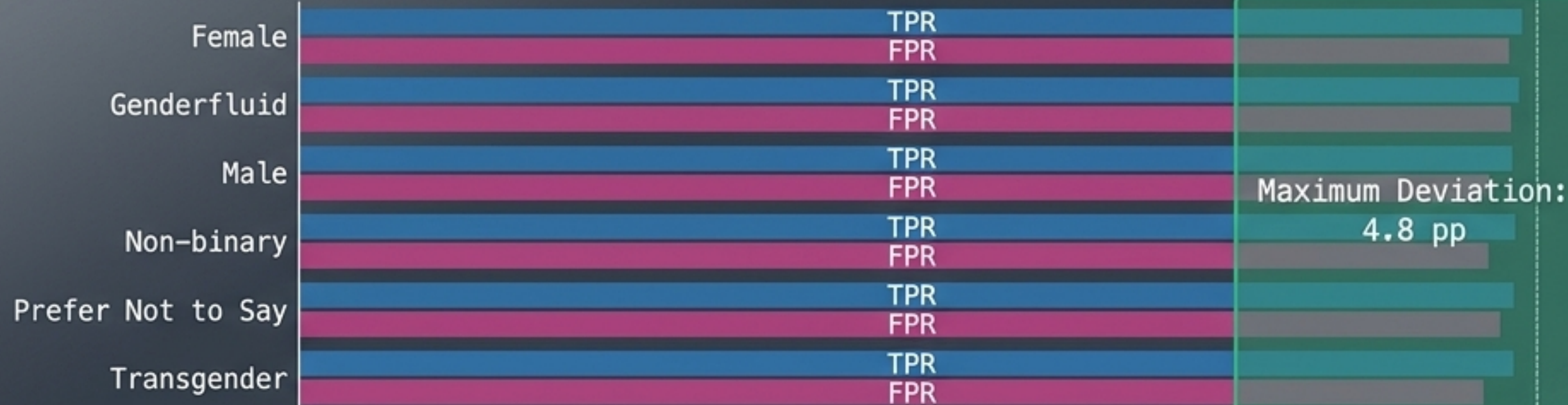
Conformal Prediction: Coverage Guarantees at Different Confidence Levels



Bayesian Uncertainty Quantification via Monte Carlo Dropout (T=100)

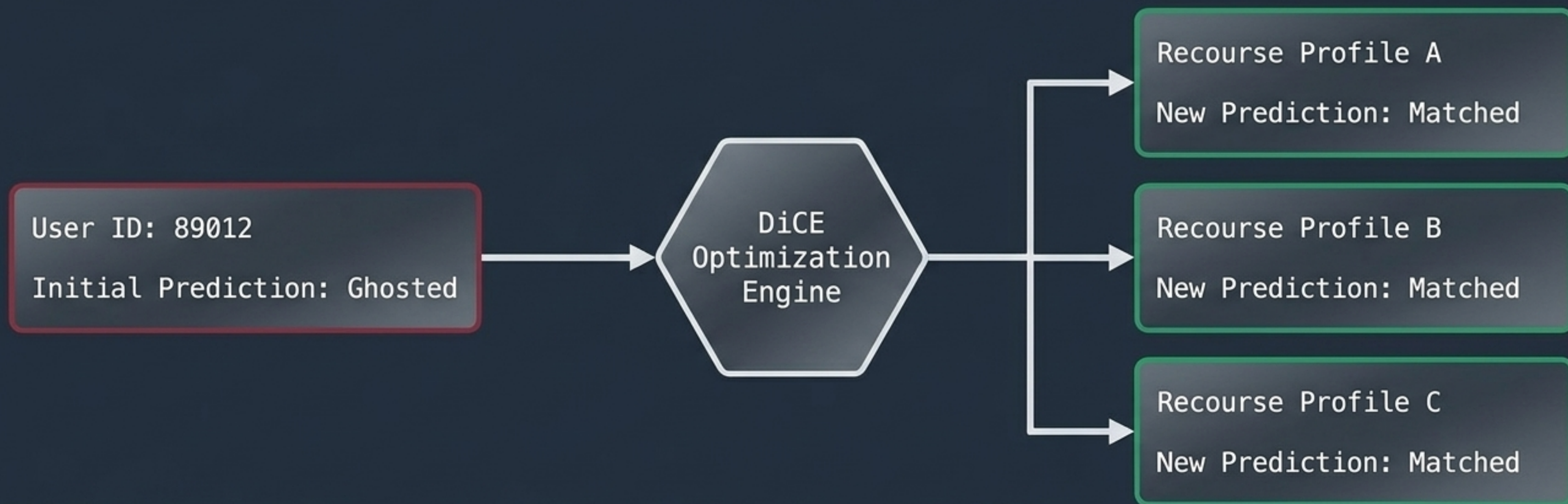


Demographic Parity Audit (Fairlearn Framework)



Strict Parity Achieved. Zero risk of algorithmic segregation.

Algorithmic Recourse: Microsoft DiCE



Minimal Actionable Interventions

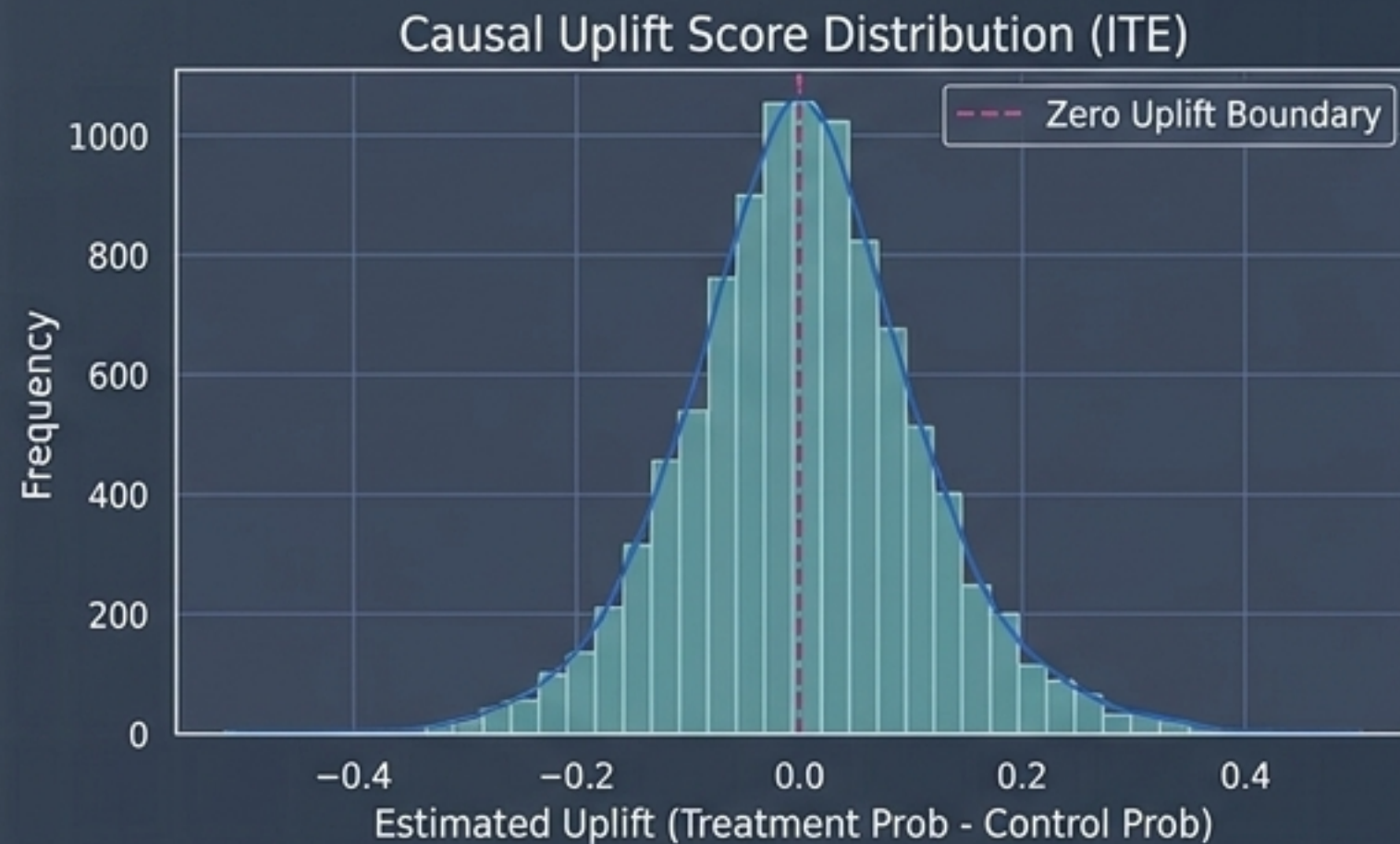
A	bio_length: +45 chars
B	profile_pics_count: +2
C	app_usage_time: +15 min



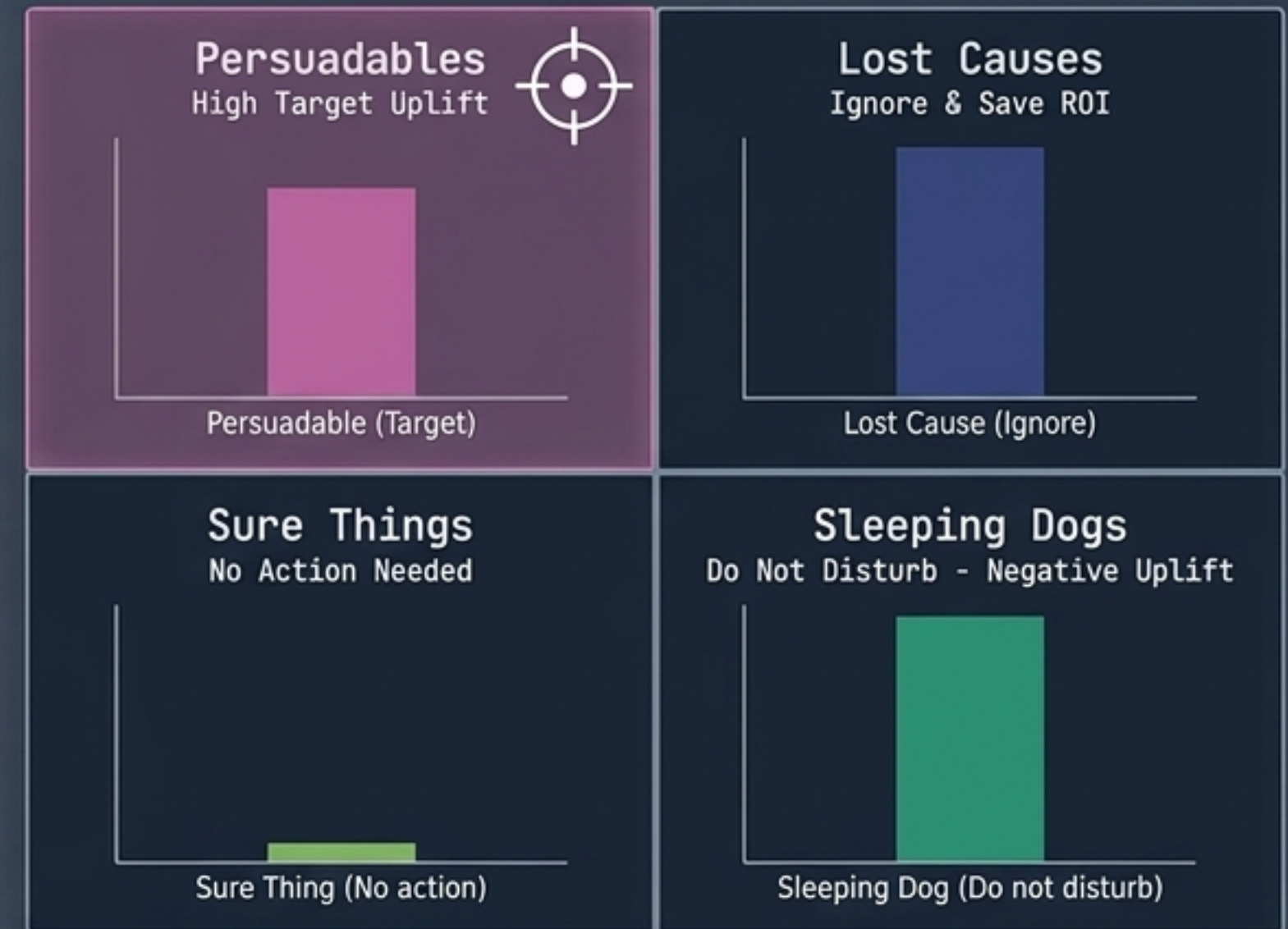
**Restricted
strictly to
mutable features.**

Causal Uplift Segment Targeting

Individual Treatment Effect (ITE)



Causal Prescriptive Marketing Matrix

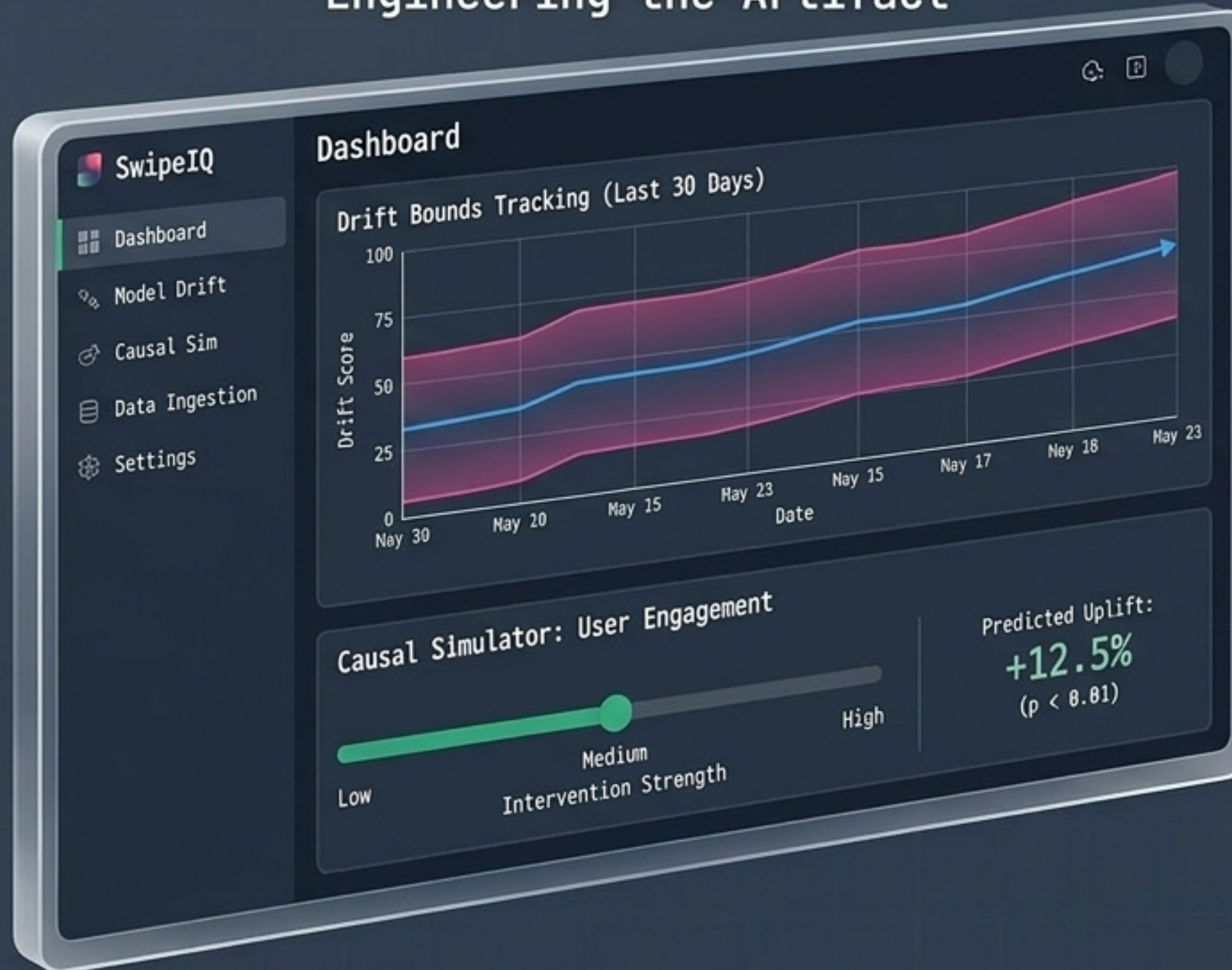


DML Average Treatment Effect (ATE) = 0.0104 (p = 0.0322)

Conclusion: Causal uplift is statistically significant but structurally minimal. Intervention ceiling reached.

SwipeIQ V2 Dashboard & Strategic Recommendations

Engineering the Artifact



The Strategic Pivot



Sunset Tabular Metrics

Mathematical proof confirms demographic and tabular features lack genuine causal signal. Predictive ceiling capped at ~60%.



Pivot to NLP Bio Analysis

Capture the true, non-linear signals of human romantic connection by analyzing semantic depth and sentiment in user text.



Integrate Behavioral Latency

Transition matching algorithms to rely on active, in-app behavioral cues and temporal response sequences.